Punit Mishra

Senior ML & AI Tech Lead — Applied Research in LLM Serving, NL-to-SQL & Multi-Tenant Systems Security

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Research & Engineering Summary

Punit Mishra is a Senior ML & AI Tech Lead at SAP CX Data & AI — 11+ years at SAP, joining through the CallidusCloud acquisition — leading a team of 13 at the intersection of large-language-model systems, natural-language-to-SQL, and secure multi-tenant architecture. His recent focus is *model optionality and accuracy*: bringing **SAP's own LLM (Nova)** into the AutoSQL NL-to-SQL stack as a self-hosted, first-class alternative to external frontier models, with an evaluation harness that measures prompt- and retrieval-level interventions — and their effect on SQL accuracy — against benchmark suites over S/4HANA and Ariba. He treats architecture as a first-class artifact, authoring ADRs (0020, 0021) that specify tenancy and cross-service auth contracts for **APEX**, SAP's **Domain Data Access Agents** platform, and the failure modes they close — cross-tenant escalation and data leakage. Alongside the applied work he sets the group's SDLC and engineering standards, owns AI ethics and compliance, and serves as a forward-deployed engineer on the first production enterprise deployment. He holds a B.S. in EECS from UC Berkeley (2012), writes production Python and Rust, and maintains several open-source systems exploring local-first LLM memory and hardware-backed security.

Research & Technical Interests

- Natural-language-to-SQL and semantic parsing over enterprise schemas
- Self-hosted multi-model LLM serving and automated evaluation
- Retrieval-augmented reranking and prompt-level accuracy interventions
- Multi-tenant SaaS identity, isolation, and cross-tenant security
- ADR-driven systems architecture and design-decision provenance
- Agent gateways and MCP-based service discovery; local-first, on-device memory for LLMs
- Evaluation rigor: benchmark design and decision-match metrics

Experience

SAP — Feb 2014 – Present • 11+ yrs

Senior ML & AI Tech Lead | CX Data & AI — *Sep 2018 – Present • San Ramon, CA*

- **Brought SAP's own LLM (Nova)** into the AutoSQL natural-language-to-SQL stack (S/4HANA + Ariba) as a first-class, self-hosted alternative to Claude — an OAuth2 serving backend with resolver-aware SAP-semantic prompts, an optional retriever–reranker, and seven benchmark cronjob lanes, including the first-ever Ariba benchmark coverage; the NL-to-SQL work ships in SAP's 2026 public and private cloud releases.
- **Drove NL-to-SQL accuracy as a measured quantity:** a date-arithmetic prompt fix moved decision-match **59.2%** → **64.8%** and reranking **50%** → **53%** — isolated and validated through the benchmark harness rather than asserted.
- **Architected and authored ADR 0021**, a hardened internal-auth contract: an X-Internal-Auth shared secret validated by constant-time compare before any tenant header is trusted, canonical tenant resolution (`app_tid` → `tenant_id`), and IAS identity alongside XSUAA — opt-in and backward-compatible across four Java and Python services, closing a real cross-tenant escalation and data-leak class by replacing trust-by-network-position with verified identity.
- **Specified APEX direct app tenancy** (ADR 0020, A2-Hybrid): turned **APEX**, SAP's **Domain Data Access Agents** platform, from manual, XSUAA-only logical tenancy into an app-owned SaaS subscription lifecycle with automatic tenant provisioning via BTP UCL formations, a CAP/cds-mtxs sidecar, a dual IAS/XSUAA token path, and a canonical tenant registry — fixing a real cross-tenant leak rooted in a shared class attribute.
- **Rewrote** the legacy Shopping Assistant's vision capabilities (Vision-DNA, object detection, visual search) as first-class agent services, exposed through a unified **Agent Gateway** and an **MCP-based service Hub** with registry and discovery, modernizing legacy code into reusable, MCP-native services.

- **Built the APEX Control Panel**, a Next.js provisioning UI on IAS OIDC PKCE with a server-side BFF (onboarding checklists, formations, agent and data-ingestion config, playground, sessions, handoff), and folded the AutoSQL eval dashboard into the Nova dashboard with HANA-backed evaluation metrics deployed on Gardener.
- **Cut** ask-a-product answer latency **~25s → ~10s** as forward-deployed engineer on the first production enterprise customer (a Fortune 500 retailer's in-store concierge), adding grounded follow-ups that stop product drift.
- **Stood up multi-model serving** on custom bare-metal H100 clusters — introducing Ray and running self-hosted Nova alongside Claude/Qwen/Granite-class models behind automated eval harnesses — giving the group model optionality, cost control, and reduced dependence on external frontier models.
- **Leads a team of 13** and sets engineering direction: guiding architecture through ADRs, owning AI ethics and compliance, defining SDLC and engineering standards, and serving as an **SAP Security Expert**.

Senior Software Engineer | Thunderbridge AI / SAP Sales Cloud — *Feb 2014 – Sep 2018 • SF Bay Area*

joined via CallidusCloud — acquired by SAP for $2.4B

- Built the **Thunderbridge AI** ML backend in Python on Dataiku with multi-tenancy running every customer workload; ported the full stack to **GKE on-prem (Anthos)** with Istio and built the CI/CD backbone (Jenkins on Kubernetes, Kaniko, Artifactory) with automated PR merges.
- Engineered multi-partition data resiliency across Oracle, Vertica, SAP HANA, and HDFS (Spark); integrated Coverity, Black Duck, and Fortify security scans per commit.

Earlier

IBM — Automation Systems Engineer — Oct 2012 – Dec 2013

Built a reporting and data-validation automation framework for a major US bank, with reusable Selenium/Java test APIs (JUnit, TestNG) adopted across the firm.

Selected Systems & Open Source

- **CoViber** — local-first memory for LLMs over the Model Context Protocol: a continuously-updated work graph, urgency-triage queue, and local semantic memory, entirely on-device with no cloud egress. *Python.*
- **Vaultic** — a hardware-backed (FIDO2/YubiKey) password manager in Rust with a single on-disk vault format shared across CLI, TUI, daemon, GUI, and MCP server. *Homebrew tap.*
- **Shield AI** — AI-driven DNS filtering with real-time ML threat scoring (Rust + TypeScript); ~15 MB resident, <1 ms lookups at 100K+ QPS, with a live demo and macOS/iOS DNS profile.
- **Railroad Arcade** — full-stack IoT (Next.js + Rust on Raspberry Pi) controlling a real HO-scale model railroad.
- **Pi Race** — a zero-dependency Rust CLI benchmarking four classic π-computation algorithms (Ramanujan, Euler/Basel, Wallis, Leibniz), with CI that cross-compiles across three platforms.

Skills & Methods

Deep Python and Rust. **Systems:** self-hosted multi-model LLM serving (Ray, bare-metal H100 clusters), FastAPI, CAP/cds-mtxs, Next.js, HANA, Gardener/Kubernetes. **Methods:** NL-to-SQL benchmark design and decision-match evaluation, retrieval reranking, prompt engineering as a measured intervention. **Security:** multi-tenant identity (IAS/XSUAA), constant-time auth contracts, ADR-driven architecture; SAP Security Expert.

Education

University of California, Berkeley — B.S. Electrical Engineering & Computer Science — 2012